Supplementary material

Direct Percutaneous Endoscopic Jejunostomy (DPEJ) & Percutaneous Endoscopic Gastrostomy with Jejunal Extension (PEG-J) Technical Success & Outcomes - A Systematic Review & Meta-Analysis

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The authors have provided this supplementary material to give readers additional information about their work.

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Supplementary material

Supplementary table 1: PRISMA checklist

Section/Topic	Item #	Checklist Item	Reported on Page #
TITLE			
Title	1	Identify the report as a systematic review *incorporating a meta-analysis.*	**1**
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: **Background:** main objectives **Methods:** data sources; study eligibility criteria, participants, and interventions; study appraisal; and synthesis methods **Results:** number of studies and participants identified; summary estimates with corresponding confidence/credible intervals and implications of findings. **Conclusion:** Implications of finds.	**4 - 5**
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known, *including mention of artificial intelligence (AI) and why a meta-analysis has been conducted.*	**6 – 7**
Objectives	4	Provide an explicit statement of questions being addressed, with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	**7**
METHODS			
Protocol and registration	5	Indicate whether a review protocol exists and if and where it can be accessed (e.g., Web address); and, if available, provide registration information, including registration number.	**8 Supplementary file table 1,2**
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	**8**
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	**8**
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	**8**
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	**8 – 9**
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	**9**
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	**9**
Risk of bias within individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	**9**
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	**9 – 10**
Planned methods of analysis	14	Describe the methods of handling data and combining results of studies for each network meta-analysis.	**10 – 11**
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	**10 - 11**
Additional analyses	16	Describe methods of additional analyses if done, indicating which were pre-specified. This may include, but not be limited to, the following: • Sensitivity analysis	**11**

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		• Subgroup analysis	
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	**11 Supplementary figure 1**
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	**11**
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	**11 - 12 Supplementary figure 2**
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: 1) simple summary data for each intervention group, and 2) effect estimates and confidence intervals using a forest plot.	**12**
Synthesis of results	21	Present results of each meta-analysis done, including confidence/credible intervals. If additional summary measures were explored (such as treatment rankings), these should also be presented.	**12 - 13**
Results of additional analyses	23	Give results of additional analyses, if done • Sensitivity analysis • Subgroup analysis	**13 - 14**
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies for the evidence base being studied (see item 15).	**14**
DISCUSSION			
Summary of evidence	24	Summarize the main findings, including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy-makers).	**15 – 19**
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review level (e.g., incomplete retrieval of identified research, reporting bias).	**19 – 20**
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	**20**
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review. This should also include information regarding whether funding has been received from manufacturers of treatments in the network and/or whether some of the authors are content experts with professional conflicts of interest that could affect use of treatments in the network.	**Disclosed in title page**

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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Supplementary table 2 MOOSE checklist

Item No	Recommendation	Reported on Page No
Reporting of background should include		
1	Problem definition	6 – 7
2	Hypothesis statement	7
3	Description of study outcome(s)	7 – 8
4	Type of exposure or intervention used	7 – 8
5	Type of study designs used	7 – 8
6	Study population	7 – 8
Reporting of search strategy should include		
7	Qualifications of searchers (eg, librarians and investigators)	8 – 9
8	Search strategy, including time period included in the synthesis and key words	8 – 9
9	Effort to include all available studies, including contact with authors	8 – 9
10	Databases and registries searched	8 – 9
11	Search software used, name and version, including special features used (eg, explosion)	8 – 9
12	Use of hand searching (eg, reference lists of obtained articles)	8 – 9
13	List of citations located and those excluded, including justification	Sup Fig 1,2
14	Method of addressing articles published in languages other than English	8 – 9
15	Method of handling abstracts and unpublished studies	8 – 9
16	Description of any contact with authors	N/A
Reporting of methods should include		
17	Description of relevance or appropriateness of studies assembled for assessing the hypothesis to be tested	8 – 9
18	Rationale for the selection and coding of data (eg, sound clinical principles or convenience)	8 – 9
19	Documentation of how data were classified and coded (eg, multiple raters, blinding and interrater reliability)	8 - 9
20	Assessment of confounding (eg, comparability of cases and controls in studies where appropriate)	9
21	Assessment of study quality, including blinding of quality assessors, stratification or regression on possible predictors of study results	9
22	Assessment of heterogeneity	10
23	Description of statistical methods (eg, complete description of fixed or random effects models, justification of whether the chosen models account for predictors of study results, dose-response models, or cumulative meta-analysis) in sufficient detail to be replicated	10
24	Provision of appropriate tables and graphics	Figs 1 - 3
Reporting of results should include		
25	Graphic summarizing individual study estimates and overall estimate	Figs 1 – 3
26	Table giving descriptive information for each study included	Tables 1 – 3
27	Results of sensitivity testing (eg, subgroup analysis)	14
28	Indication of statistical uncertainty of findings	14

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Item No	Recommendation	Reported on Page No
Reporting of discussion should include		
29	Quantitative assessment of bias (eg, publication bias)	14, Supp Fig 3
30	Justification for exclusion (eg, exclusion of non-English language citations)	15 – 19
31	Assessment of quality of included studies	15 – 19
Reporting of conclusions should include		
32	Consideration of alternative explanations for observed results	19 – 20
33	Generalization of the conclusions (ie, appropriate for the data presented and within the domain of the literature review)	19 - 20
34	Guidelines for future research	20
35	Disclosure of funding source	Title page

From: Stroup DF, Berlin JA, Morton SC et al. for the Meta-analysis Of Observational Studies in Epidemiology (MOOSE) Group. Meta-analysis of Observational Studies in Epidemiology. A Proposal for Reporting. JAMA. 2000;283(15):2008-2012. doi: 10.1001/jama.283.15.2008

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Supplementary table 3 Search Strategy used in June, 2021

Literature Database	Search items	Items found
Embase	"1 percutaneous endoscopic gastrostomy/ (6,289) 2 jejunostomy/ (5,858) 3 jejunostomy tube/ (818) 4 direct percutaneous endoscopic jejunostomy.dq. (24) 5 percutaneous jejunostomy.dq. (11) 6 (percutaneous adj3 (jejunal or jejunostomy)).ab,kw,ti. (462) 7 (PEG-J or DPEJ or PEJ).ab,kw,ti. (529) 8 1 or 2 or 3 or 4 or 5 or 6 or 7 (12,688) 9 exp treatment outcome/ (1,765,771) 10 (adverse or outcome$ or complication$ or fail$ or response$ or result$ or analy$).ab,kw,ti. (19,786,553) 11 9 or 10 (20,137,134) 12 (percutaneous adj3 (jejunal or jejunostomy)).kw,ti. (255) 13 (PEG-J or DPEJ or PEJ).kw,ti. (112) 14 12 or 13 (333) 15 8 and 11 and 14 (247) 16 exp case report/ (2,593,602) 17 case report$.ab,kw,ti. (527,508) 18 16 or 17 (2,664,743) 19 15 not 18 (204) 20 limit 19 to (article or article in press or ""review"") (136) 21 exp juvenile/ (3,561,916) 22 exp adult/ (8,924,939) 23 21 not 22 (2,275,129) 24 20 not 23 (134) 25 exp animal/ (26,870,464) 26 exp human/ (22,100,067) 27 25 not 26 (4,770,397) 28 24 not 27 (132) 29 limit 28 to english language (110)"	110
MEDLINE	"1 Jejunostomy/ (2,809)2 (percutaneous adj3 (jejunal or jejunostomy)).ab,kf,ti. (287) 3 (PEG-J or DPEJ or PEJ).ab,kf,ti. (242) 4 1 or 2 or 3 (3,039) 5 exp Outcome Assessment, Health Care/ (1,175,749) 6 (adverse or outcome$ or complication$ or fail$ or response$ or result$ or analy$).ab,kf,ti. (15,332,789) 7 5 or 6 (15,564,232) 8 (percutaneous adj3 (jejunal or jejunostomy)).kf,ti. (172) 9 (PEG-J or DPEJ or PEJ).kf,ti. (57) 10 8 or 9 (210) 11 4 and 7 and 10 (136) 12 exp Case Reports/ (2,160,120) 13 case report$.ab,kf,ti. (416,167) 14 12 or 13 (2,250,586) 15 11 not 14 (103) 16 exp Infant/ (1,159,035) 17 exp Child/ (2,072,603) 18 exp Adolescent/ (2,072,603) 19 16 or 17 or 18 (3,634,123) 20 exp Adult/ (7,388,775) 21 19 not 20 (1,917,141) 22 15 not 21 (102) 23 exp Animals/ (23,872,866) 24 exp Humans/ (19,076,340) 25 23 not 24 (4,796,526) 26 22 not 25 (101) 27 limit 26 to english language (94)"	94
CENTRAL	"1 jejunostomy.sh. (82) 2 (percutaneous adj3 (jejunal or jejunostomy)).ab,kw,ti. (16) 3 (PEG-J or DPEJ or PEJ).ab,kw,ti. (28) 4 1 or 2 or 3 (117) 5 outcome assessment, health care.sh. (8,452) 6 (adverse or outcome$ or complication$ or fail$ or response$ or result$ or analy$).ab,kw,ti. (1,371,162) 7 5 or 6 (1,371,941) 8 4 and 7 (101) 9 (percutaneous adj3 (jejunal or jejunostomy)).kw,ti. (3) 10 (PEG-J or DPEJ or PEJ).kw,ti. (3) 11 9 or 10 (6) 12 8 and 11 (5) 13 conference.pt. (3) 14 conference abstract.pt. (16,876) 15 13 or 14 (16,879) 16 12 not 15 (4) 17 chinese.lg. (36,990) 18 16 not 17 (3)"	3
Scopus	((TITLE-ABS-KEY (jejunostomy OR "percutaneous endoscopic gastrostomy" OR "jejunostomy tube") OR TITLE-ABS-KEY ("direct percutaneous endoscopic jejunostomy" OR "percutaneous jejunostomy") OR TITLE-ABSKEY (percutaneous W/3 (jejunal OR jejunostomy)) OR TITLE-ABS-KEY (peg-j OR dpej OR pej))) AND ((TITLE-ABS-KEY ("outcome assessment, health care" OR "treatment outcome") OR TITLE-ABS-KEY (adverse OR outcome* OR complication* OR fail* OR response* OR result* OR analy*))) AND ((TITLE (percutaneous W/3 (jejunal OR jejunostomy)) OR KEY (percutaneous W/3 (jejunal OR jejunostomy)) OR TITLE (peg-j OR dpej OR pej) OR KEY (peg-j OR dpej OR pej))) AND (EXCLUDE (DOCTYPE, "cp")) AND (LIMIT-TO (EXACTKEYWORD, "Human") OR LIMIT-TO (EXACTKEYWORD, "Humans")) AND (LIMIT-TO LANGUAGE, "English")) AND (EXCLUDE (EXACTKEYWORD, "Case Report") OR EXCLUDE (EXACTKEYWORD, "Adolescent") OR EXCLUDE (EXACTKEYWORD , "Child") OR EXCLUDE (EXACTKEYWORD, "Child, Preschool")) View less	126
Web of Science	"#1 ((TS=(jejunostomy OR ""percutaneous endoscopic gastrostomy"" OR ""jujenostomy tube"" OR ""direct percutaneous endoscopic jejunostomy"" OR ""percutaneous jejunostomy"")) OR TS=(percutaneous NEAR/3 (jejunal or jejunostomy))) OR TS=(""PEG-J"" OR DPEJ OR PEJ) (7284) #2 (TS=(""outcome assessment, health care"" OR ""treatment outcome"")) OR TS=(adverse OR outcome* OR complication* OR fail* OR response* OR result* OR analy*) (22766970) #3 (((TI=(percutaneous NEAR/3 (jejunal OR jejunostomy))) OR KP=(percutaneous NEAR/3 (jejunal OR jejunostomy))) OR TI=(""PEG-J"" OR DPEJ OR PEJ)) OR KP=(""PEG-J"" OR DPEJ OR PEJ) (356) #4 ((#1) AND #2) AND #3 (161) #5 (TS=(""case report"" OR ""case reports"")) OR TS=(conference* OR ""conference abstract"" OR ""conference abstracts"") OR TS=(meeting* OR symposium*) (601820) #6 #6 (#4) NOT #5 (158) #7 (TS=(infant* OR child* OR adolescen* OR juvenile*)) NOT TS=(adult*) (1620335) #8 (#6) NOT #7 (150) #9 (TS=(animal*)) NOT TS=(human*) (828554) #10 (#8) NOT #9 (150) #11 (#8) NOT #9 and Meeting Abstracts or Proceedings Papers (Exclude – Document Types) (118)"	118

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Supplementary material

Total		451
Duplicates		250

Supplementary figure 1 - Quality assessment of included studies using the Qumseya scale

Author/Year	Total	Interpretation
Ponsky 1984	5	Low
Shike 1987	5	Low
Kaplan 1989	5	Low
Shike 1991	8	High
Mellert 1994	8	High
Shike 1996	8	High
Cococcia 2000	6	Moderate
Rumalla 2000	5	Low
Barrera 2001	7	Moderate
Shetzline 2001	7	Moderate
Varadarajulu 2003	8	High
Bueno JT 2003	7	Moderate
Maple 2005	8	High
Del Piano M 2008	7	Moderate
Mackenzie 2008	7	Moderate
Panagiotakis 2008	6	Moderate
Moran 2009	6	Moderate
Aktas 2012	6	Moderate
Song 2012	8	High
Toussaint 2012	6	Moderate
Lim 2015	8	High
Velázquez-Aviña 2015	7	Moderate
Al-Bawardy 2016	7	Moderate
Bernardes 2017	6	Moderate
Strong 2017	5	Low
Kirstein 2018	6	Moderate
Ridtitid 2018	7	Moderate
Simoes 2018	6	Moderate
Nishiwaki 2021	9	High

Interpretation: <6 = Low, 6 – 7 = Moderate and >8 = High

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